

Provenance, Not Behaviour: A Serialisation Artifact in Edge-IIoTset and a Leakage-Free Benchmark for Precision-Agriculture Intrusion Detection

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Abstract

Edge-IIoTset is the reference benchmark for machine-learning intrusion detection in the industrial Internet of Things, and results reported on it cluster above 99%. We show that much of that performance is not intrusion detection. The preprocessing recipe distributed with the dataset instructs researchers to one-hot encode seven categorical columns. Four of them separate attack from normal traffic with an accuracy of 1.0000 on their own, through the spelling of the placeholder written for an absent protocol field: the string 0 in the normal-traffic branch of the dataset build against 0.0 in the attack branch. The label is recoverable from a serialisation artifact encoding file provenance, with no network behaviour modelled, and separates every row of both curated subsets. Under 5-fold $\times$ 3-repeat cross-validation, five of six standard classifiers attain exactly 1.0000 ± 0.0000 accuracy and the sixth attains 0.99998. Under a corrected protocol, naive Bayes falls by 0.3005 macro-F1 and the strongest model settles at 0.9503 ± 0.0011 . Label, ordinal and frequency encoding leak identically. Because the curated subsets also lack Modbus and per-device identity, we rebuild the benchmark from the raw captures under uniform parsing, producing **AgriEdge**: 1,276,122 rows, five devices with full attribution, and no column separating the classes above 0.0288. A leave-one-device-out sweep locates the generalisation boundary at the perception/actuation layer, where random forest falls from 0.9988 to 0.5083 balanced accuracy. Non-IID federated partitioning costs at most 0.0037 macro-F1, but a 20-round LoRaWAN training run costs 4.6 hours of uplink.

Keywords: intrusion detection, industrial IoT, precision agriculture, data

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1. Introduction

Precision agriculture has become one of the more consequential deployments of the industrial Internet of Things (IIoT). Soil-moisture probes, pH sensors, water-level floats and climate stations feed telemetry to edge gateways over lightweight publish/subscribe protocols, and those gateways actuate irrigation pumps and valves over fieldbus protocols such as Modbus TCP. The consequences of compromise are physical and immediate: a falsified soil-moisture reading can drive a controller to flood a field, and a denial-of-service attack against a broker can stop an irrigation trigger firing in the narrow window that matters. Unlike an enterprise network, a farm has no on-site security staff, intermittent connectivity, and gateways sharing a power budget with the machinery they control.

These constraints have made machine-learning intrusion detection attractive, and the Edge-IIoTset dataset of Ferrag et al. [1] has become the dominant vehicle for evaluating it. On paper it is unusually well-suited: captured from a physical seven-layer testbed with real sensors, spanning fourteen attack classes across MQTT, Modbus, HTTP and DNS, and explicitly partitioned for both centralised and federated learning. It is also convenient at 1.2 GB, and ships with a `Readme.txt` giving a step-by-step preprocessing recipe.

The results reported on it are uniformly extraordinary—99.27% accuracy with 99.21% F1 [2], 99.94% accuracy [3]—and such numbers are ordinarily read as evidence that the problem is close to solved and that attention should move to deployment concerns. This paper argues they should be read as a warning. We set out to build a precision-agriculture intrusion detector on Edge-IIoTset following the standard pipeline, and audited the features the dataset’s own documentation instructs researchers to construct. The recipe manufactures a near-perfect label proxy out of a serialisation artifact.

1.1. The artifact in brief

Edge-IIoTset encodes an absent protocol field as a zero rather than as a null—reasonable, since a packet carrying no MQTT layer has no MQTT topic. The normal-traffic captures and the attack captures were, however, parsed separately and concatenated afterwards, and the two branches serialised that zero differently: as the string 0 in one and as the string 0.0 in

the other. Step 5 of the distributed recipe then applies `pandas.get_dummies` to seven columns containing these placeholders. Dummy encoding treats 0 and 0.0 as distinct tokens, so the resulting binary features encode *which file a row came from*. Because normal traffic and attack traffic came from different files, file provenance is the label (Figure 1).

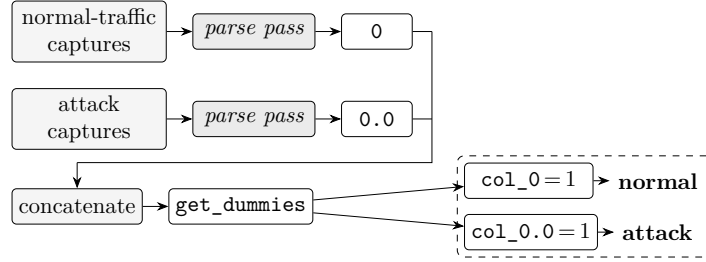


Figure 1: How file provenance becomes the label. The two captures were parsed in separate passes that serialise an absent field differently, and dummy encoding turns the two spellings into indicator columns. Since the files partition the classes, either column reproduces the label. No step involves network behaviour.

The consequence is that four of the seven columns in the recipe recover the binary label with an accuracy of 1.0000 in isolation. No model that consumes those features is doing intrusion detection; it is reading a build artifact.

1.2. Contributions

1. **We identify a label-leakage mechanism in the canonical Edge-IIoTset recipe**, localise it to placeholder serialisation, and quantify it with token purity, single-column held-out accuracy and normalised mutual information. Four columns reach a separation rate of 1.0000 (Section 5.1).
2. **We measure what the artifact is worth.** Under 5-fold $\times$ 3-repeat cross-validation five of six standard classifiers reach exactly 1.0000 ± 0.0000 accuracy in every fold and the sixth reaches 0.99998; under correction the same models span 0.6995 to 0.9503 macro-F1. A 1D-CNN and a deep MLP reach 1.0000 on a single stratified split and lose more than 0.21 macro-F1 under correction (Section 5.2).
3. **We show the leak is encoding-agnostic.** One-hot, label, ordinal and frequency encoding all yield identical 1.0000 single-column accuracy, each being a bijection on the token set. The exposed population is every study that treated these columns as categorical, not only those copying the recipe (Section 5.2.3).

4. **We rebuild the benchmark.** The curated subsets cannot support agricultural research—Modbus effectively absent, device identity stripped—so we reconstruct from the raw captures under uniform parsing (Sections 5.3 and 5.4).
5. **We locate the generalisation boundary.** A leave-one-device-out sweep over all five devices shows transfer between perception sensors is near-free—every model but naive Bayes scores 0.9880–0.9999 balanced accuracy when soil moisture, water level or pH is withheld, and 0.9106–0.9385 for temperature and humidity—while transfer across the perception/actuation layer collapses to 0.4877–0.5083, straddling the trivial baseline, for two of the six (Section 5.5).
6. **We show the federated literature is optimising the wrong constraint.** Non-IID partitioning costs at most 0.0037 macro-F1, while training a 28,450-parameter model over LoRaWAN costs 4.6 hours of uplink; deployment cost varies $342\times$ in latency and $171\times$ in model size across models whose random-split accuracies are indistinguishable (Sections 5.6 and 5.7).
7. **We release the complete pipeline**—audit tooling, benchmark construction and experiments—so the audit can be applied to other datasets (Section 8).¹

2. Background

2.1. Precision agriculture as a layered IIoT system

The precision-agriculture deployments that Edge-IIoTset’s testbed models decompose into three layers, and the security properties of each differ sharply.

The perception layer comprises the sensors observing the physical field: soil moisture, pH, water level, temperature and humidity. Their traffic is low-rate, periodic and semantically narrow—a soil-moisture probe emits a scalar at a fixed interval. That regularity makes anomaly detection tractable here, and makes the layer a target: an adversary who alters a reading controls the actuation logic downstream without touching the actuator.

The network layer carries that telemetry. MQTT dominates sensor reporting because its publish/subscribe model and small header suit constrained devices, while Modbus TCP remains the lingua franca for actuation. Neither authenticates nor encrypts by default in typical farm deployments.

¹<https://github.com/MostafaGalal1/agriedge>

The edge system layer performs local processing. Agricultural sites are often served by cellular or LoRaWAN backhaul with severely constrained uplink, so detection must run on the gateway rather than in the cloud, where the tension between detection quality and resource budget binds.

2.2. The Edge-IIoTset dataset

Edge-IIoTset [1] was captured from a physical testbed spanning IoT sensors, edge gateways, SCADA components and adversary hosts. It is distributed in three parts: raw per-device normal-traffic captures (CSV and PCAP), raw per-attack captures, and two curated subsets—`ML-EdgeIIoT-dataset.csv` (157,800 rows) and `DNN-EdgeIIoT-dataset.csv` (2,219,201 rows)—intended for traditional machine learning and deep learning respectively. Both curated subsets carry 63 columns: 61 protocol features, a binary `Attack_label`, and a 15-class `Attack_type`.

Nearly all published work uses the curated subsets, and follows the `Readme.txt` recipe: drop fifteen identifier and payload columns, drop nulls and duplicates, and dummy-encode seven categorical columns. That recipe is the object of our audit.

2.3. Related work

Performance on Edge-IIoTset. The literature is dense and its results are tightly clustered near ceiling. Reported figures include 99.27% accuracy with 99.21% F1 across four benchmark datasets [2] and 99.94% accuracy with a confusion matrix containing no errors of either kind [3]. Architectures range from autoencoder-based lightweight feature learning [3] to hybrid CNN-DNN models and TinyML deployments for energy-aware edge inference [4].

Emerging scepticism. A parallel literature questions whether these numbers mean what they appear to. Hakim et al. [5] train four architectures on one IIoT dataset and evaluate them without retraining on two structurally distinct ones, finding reliance on coarse port-category shortcuts appearing $96\text{--}435\times$ more often in source-domain attacks. A dataset-centric review [6] catalogues evaluation biases across IoT/IIoT intrusion detection—undocumented preprocessing, hidden leakage, inconsistent validation—and notes that no study reports k -fold cross-validation on Edge-IIoTset, relying on 80/20 splits. Work on cross-domain benchmarking [7] and dataset-centric federated evaluation [8] points the same way.

Benchmark integrity as a research programme. A distinct line of work audits intrusion detection benchmarks directly rather than proposing models for them. Engelen et al. [9] reverse-engineer CICIDS2017 and find labelling and traffic-generation faults severe enough to change reported conclusions. Flood et al. [10] generalise this into a catalogue of six recurring *data design smells* obtained by manual analysis of seven highly-cited benchmark datasets. Two of the six sit near our finding: *highly dependent features*, covering features of outsized importance that are unrelated to the underlying mechanism of an attack, and *wrong label*, covering mislabelled data produced by inaccurate ground truth in a generation testbed. Arp et al. [11] survey thirty top-tier security papers and codify the recurring methodological pitfalls of learning-based security, spurious correlation among them.

This paper sits inside that programme rather than alongside it, and two things distinguish its contribution. First, Edge-IIoTset is not among the seven datasets Flood et al. examine, and we are not aware of any comparable audit of it. Second, the fault sits where the catalogue does not look. All six smells locate the problem in the testbed—in how traffic was generated, configured or labelled. Here the captures are sound, the ground truth is accurate and every label is correct; the artifact enters downstream, in the pipeline that serialises the captures into a feature table, because the two label classes were parsed in separate passes that disagree about how to spell an absent value. So it is not *wrong label*, since no label is wrong, and not quite *highly dependent features* either: the features documented under that heading are genuine network properties—destination port, packet size, flow duration—whereas the token carrying our signal is the placeholder written where a field was *absent*, and its spelling records which file the row came from. The dependency is perfect rather than strong, it originates downstream of capture, and it is invisible to any inspection that casts the column to a numeric type first.

Two results are closer still to ours. Bouke and Abdullah [12] study *pattern leakage* introduced during preprocessing rather than present in the raw capture, and show empirically that it inflates reported reliability—establishing the general phenomenon that this paper instantiates in a specific dataset with a specific mechanism. Kostas et al. [13] show that individual-packet features let models memorise artifacts that do not survive transfer to another network, which is the failure mode our leave-one-device-out sweep exhibits at a layer boundary (Section 5.5). Neither reports a label-recovering artifact in Edge-IIoTset, and to our knowledge no published work does.

Our position relative to this work. Two literatures meet here. The IIoT-specific sceptical work establishes *that* generalisation fails without localising a mechanism; the benchmark-integrity work localises mechanisms precisely, but in classical network-intrusion datasets. This paper applies the latter’s auditing stance to the dataset the former is built on, and finds a mechanism both stronger than those previously catalogued—it recovers the label alone, at accuracy 1.0000—and different in kind, since the discriminating value encodes file provenance rather than any property of the traffic.

Federated learning for agricultural IIoT. Federated approaches have been proposed specifically for agricultural intrusion detection, including secure federated deep reinforcement learning [14]. These evaluations typically partition data uniformly at random, producing independent and identically distributed (IID) clients. Section 5.6 shows why that choice matters.

3. Threat Model

We consider an adversary with network access to a farm’s IIoT segment, obtained through a compromised gateway, an exposed cellular modem, or physical proximity to an unsecured wireless link. The adversary cannot compromise the sensors’ physical measurements but can observe, inject and modify network traffic. We assume the detector runs on the edge gateway and that the gateway is trusted; detecting a compromised gateway is out of scope.

The attack classes in Edge-IIoTset map onto agricultural consequences directly. **Man-in-the-middle** (ARP and DNS spoofing) alters telemetry in flight—the falsified-moisture scenario. **Denial of service** (TCP SYN, UDP, ICMP and HTTP floods) severs the telemetry path, preventing irrigation triggers. **Reconnaissance** precedes targeted attacks, **injection and application attacks** target the interfaces through which operators configure irrigation, and **backdoor and ransomware** establish persistence—ransomware against an irrigation controller carrying an unusually compressed decision timeline given crop water stress.

Two of these classes are severely under-represented in the dataset—MITM at 1,229 rows and OS fingerprinting at 1,001—despite MITM being the most consequential for the falsified-telemetry scenario. We preserve this natural rarity rather than resampling it away, and report per-class recall accordingly.

4. Methodology

4.1. Auditing for label leakage

Our audit asks one question per categorical column: *how much of a classifier’s apparent skill can this column supply on its own, without any network behaviour being modelled?* We compute three measures so that no single statistic carries the argument.

Token purity. For each distinct token we count occurrences under each label; a token is *pure* if it occurs under exactly one. The **separation rate** is the fraction of rows covered by pure tokens, and a column at 1.0 is a relabelling of the target.

Single-column held-out accuracy. We fit a decision tree to the one-hot encoding of that column alone under a stratified 80/20 split and report test accuracy, measuring directly what a model can extract from it.

Normalised mutual information. between the column’s tokens and the label, which is estimator-free and scale-free.

We additionally run a mechanism-specific probe. For each column containing both 0 and 0.0 we tabulate the label distribution of each spelling. If each occurs under exactly one label, the column carries a **provenance marker**: the spelling of a value denoting *absence*—which has no network semantics whatsoever—determines the label with certainty.

The columns audited are exactly the seven `Readme.txt` Step 5 instructs researchers to dummy-encode: `http.request.method`, `http.referer`, `http.request.version`, `dns.qry.name.len`, `mqtt.conack.flags`, `mqtt.protoname` and `mqtt.topic`. Critically, we read all columns as strings—pandas’ type inference silently normalises 0 and 0.0 to the same float, destroying the artifact before it can be observed, which is a plausible reason it has gone unreported.

4.2. The two preprocessing protocols

Protocol A (as distributed). reproduces `Readme.txt` Steps 4 and 5 verbatim: drop the fifteen listed columns, drop rows with nulls, drop duplicates, dummy-encode the seven listed columns, and coerce the remainder to numeric.

Protocol B (corrected). applies four changes:

1. **Placeholder canonicalisation.** Every spelling denoting absence (0, 0.0, empty, nan, None, null) is collapsed to a single sentinel before encoding. This removes the provenance signal while preserving the field’s semantic content: *this protocol layer was not present in this packet*.
2. **Deduplication before splitting.** Exact duplicates spanning a train/test boundary are memorised rather than generalised.
3. **Structural rather than identity encoding of high-cardinality strings.** Columns such as `http.request.version` contain whole injected request lines on the attack side. One-hot encoding these lets a model memorise literal payload strings, which neither transfers to another network nor survives a single changed byte. We replace token identity with structural properties—length, character-class ratios, Shannon entropy, and presence of injection-associated syntax.
4. **Post-hoc removal of residual separators.** Any column still separating the classes above a configurable threshold (default 0.999) after the above is dropped.

4.3. Constructing the AgriEdge benchmark

The curated subsets cannot support an agricultural study (Section 5.3). We therefore rebuild from the raw per-device captures, which retain full device attribution. All raw captures share a byte-identical 63-column header, which makes uniform construction possible.

Two properties are enforced. **Uniform parsing:** every capture—normal and attack alike—is read with identical dtype handling and identical placeholder canonicalisation applied across *all* columns. Because the artifact arises from concatenating differently-parsed frames, parsing uniformly makes it structurally impossible rather than merely correcting it after the fact. **Preserved device attribution:** each row records the device or attack capture it came from, which is what makes non-IID federated partitioning by farm possible, and which the curated subsets discard.

Sampling is per-chunk at a constant rate rather than by prefix, which is unbiased with respect to position in the capture session. Normal devices are capped at 150,000 rows each and attack classes at 60,000; because sampling is per chunk, a class may overshoot its cap by less than one chunk. Classes smaller than the cap are taken whole, preserving the natural rarity of MITM and fingerprinting. Host identifiers and raw payload columns are dropped at build time so no downstream protocol can reintroduce them.

4.4. Evaluation protocols

We evaluate under two splitting regimes. **Random stratified** (80/20) is the convention in the literature and provides comparability. **Leave-one-device-out (LODO)** withholds all normal traffic from one sensor type and tests on it, with 20% of attack traffic. This models the deployment reality that a farm adds a sensor the detector was never trained on. It is a genuine distribution shift, and the closest within-dataset analogue of the cross-network evaluation that Hakim et al. [5] argue is missing from the literature.

4.5. Model suite

We use six estimators chosen to mirror those most frequently reported on Edge-IIoTset: decision tree, random forest, histogram gradient boosting, logistic regression, Gaussian naive Bayes, and a two-hidden-layer MLP (64, 32). Every factory returns a fresh estimator; none is shared between protocols, folds, or federated clients. All randomness is seeded.

The deliberate inclusion of weak models is methodological. If a classifier that cannot represent the decision boundary nonetheless achieves perfect accuracy, the features—not the classifier—are doing the work. Gaussian naive Bayes serves as this canary throughout.

4.6. Federated and edge-cost evaluation

Federated experiments use FedAvg [15] over a compact MLP sized for a gateway, comparing three client constructions: IID (the prior-work baseline), one client per sensor type, and one client per farm, where farms group devices as {soil moisture, water level}, {temperature-humidity, pH}, and {Modbus}. Clients exchange only parameters.

Edge cost is measured as serialised model size, parameter count, and inference latency reported as median and 95th percentile over 30 timed single-sample batches. We report the tail because a late verdict is a missed actuation window. Communication cost is computed against representative rural backhaul profiles: LoRaWAN SF7 (5.47 kbit/s), NB-IoT (250), LTE Cat-M1 (1,000), and rural 4G (5,000).

5. Results

5.1. Four columns recover the label perfectly

Table 1 reports the audit of the seven columns named in the distributed recipe, on the ML subset (157,800 rows; 24,301 normal, 133,499 attack).

Table 1: Leakage audit of the columns `Readme.txt` Step 5 instructs researchers to dummy-encode (ML subset).

Column	Tokens	Pure normal	Pure attack	Rows separated	Separation rate	1-col. accuracy	NMI
<code>dns.qry.name.len</code>	8	6	2	157,800	1.0000	1.0000	0.9859
<code>mqtt.conack.flags</code>	3	2	1	157,800	1.0000	1.0000	0.9642
<code>mqtt.protoname</code>	3	2	1	157,800	1.0000	1.0000	0.9649
<code>mqtt.topic</code>	3	2	1	157,800	1.0000	1.0000	0.9650
<code>http.request.version</code>	8	0	7	62,472	0.3959	0.8460	0.1378
<code>http.request.method</code>	6	0	5	61,258	0.3882	0.8460	0.1348
<code>http.referer</code>	4	0	3	30,689	0.1945	0.8460	0.0783

Four columns separate every row in the dataset. A decision tree given one of these columns and nothing else achieves perfect held-out accuracy. Normalised mutual information with the label reaches 0.9859.

Table 2 isolates the mechanism.

Table 2: Label distribution of the two spellings of the zero placeholder (ML subset).

Column	'0'		'0.0'	
	Normal	Attack	Normal	Attack
<code>dns.qry.name.len</code>	24,272	0	0	133,272
<code>mqtt.conack.flags</code>	23,012	0	0	133,499
<code>mqtt.protoname</code>	23,051	0	0	133,499
<code>mqtt.topic</code>	23,055	0	0	133,499
<code>http.request.method</code>	0	54,062	24,301	72,241
<code>http.referer</code>	0	30,399	24,301	102,810
<code>http.request.version</code>	0	55,276	24,301	71,027

The pattern is unambiguous. In the four fully-separating columns, '0' never appears in an attack row and '0.0' never appears in a normal row. The three HTTP columns show the same artifact with the polarity reversed—'0' appears only in attack rows—separating 19–40% of the data.

This is a build artifact, not a network phenomenon. Both tokens denote the same thing: the protocol layer was absent. Their distinction records only which parsing branch wrote the row. We additionally measured duplicate rows, which inflate accuracy under random splitting: 814 exact duplicates (0.52%), all with consistent labels.

The artifact is not confined to the small subset. Because deep-learning studies use the larger `DNN-EdgeIIoT-dataset.csv`, we repeated the audit there. Table 3 shows the identical mechanism at fourteen times the scale—and on a subset whose class balance is *inverted* relative to the ML subset (1,615,643

normal against 603,558 attack), ruling out any explanation contingent on which class is the majority.

Table 3: The same four columns on the 2,219,201-row DNN subset.

	Separation rate	1-col. accuracy	NMI	'0' normal / attack	'0.0' normal / attack
<code>dns.qry.name.len</code>	1.0000	1.0000	0.9927	1,613,798 / 0	0 / 603,331
<code>mqtt.conack.flags</code>	1.0000	1.0000	0.8879	1,532,586 / 0	0 / 603,558
<code>mqtt.protoname</code>	1.0000	1.0000	0.8881	1,532,608 / 0	0 / 603,558
<code>mqtt.topic</code>	1.0000	1.0000	0.8881	1,532,627 / 0	0 / 603,558

Every one of 2,219,201 rows is separated, by each of four columns independently. Both branches of the Edge-IIoTset literature—the classical-ML branch using the ML subset and the deep-learning branch using the DNN subset—are therefore exposed to the same artifact.

5.2. What the artifact is worth

Because a single split cannot distinguish an effect from split variance—and because the review we cite [6] notes that no published study reports k -fold on this dataset—we evaluate under 5-fold $\times$ 3-repeat stratified cross-validation, reporting mean and 95% interval over all fifteen folds (Table 4).

Table 4: Binary classification under the distributed recipe (A) versus the corrected protocol (B). ML subset, 15 folds, mean $\pm$ 95% CI half-width. Five of the six models score exactly 1.0000 ± 0.0000 under the recipe; the MLP row is reported to five decimals because it does not, and rounding it to four would conceal that.

Model	Accuracy (A)	Accuracy (B)	Macro-F1 (A)	Macro-F1 (B)	Δ F1
Hist. gradient boosting	1.0000 $\pm$ 0.0000	0.9752 $\pm$ 0.0005	1.0000 $\pm$ 0.0000	0.9503 $\pm$ 0.0011	0.0497
Random forest	1.0000 $\pm$ 0.0000	0.9658 $\pm$ 0.0005	1.0000 $\pm$ 0.0000	0.9343 $\pm$ 0.0009	0.0657
Decision tree	1.0000 $\pm$ 0.0000	0.9610 $\pm$ 0.0005	1.0000 $\pm$ 0.0000	0.9271 $\pm$ 0.0009	0.0729
MLP	0.99998 $\pm$ 0.00001	0.9095 $\pm$ 0.0008	0.99996 $\pm$ 0.00003	0.7858 $\pm$ 0.0023	0.2142
Logistic regression	1.0000 $\pm$ 0.0000	0.8993 $\pm$ 0.0008	1.0000 $\pm$ 0.0000	0.7567 $\pm$ 0.0024	0.2433
Gaussian naive Bayes	1.0000 $\pm$ 0.0000	0.8742 $\pm$ 0.0125	1.0000 $\pm$ 0.0000	0.6995 $\pm$ 0.0089	0.3005

Three observations.

Five of the six models attain exactly 1.0000 ± 0.0000 , in every one of fifteen folds. Not 0.999, and not on a lucky split: zero variance across fifteen independent stratified partitions, for five structurally different learners. The sixth, the small (64, 32) MLP, averages 0.99998—fewer than one misclassified row per fold out of 31,560, and the only model in the suite whose recipe score is not bit-exact. We report it unrounded rather than let it round up to the same 1.0000 as the rest, since a paper about numbers

that were never checked closely should survive being checked closely itself. Perfect or near-perfect invariant separation of a real network-traffic capture is not a plausible outcome of intrusion detection. It is the signature of a feature that *is* the label.

The canary fires. Gaussian naive Bayes assumes conditional independence of features given the class and cannot represent the interactions a genuine intrusion-detection boundary requires. Its perfect, zero-variance score under the distributed recipe is the clearest evidence available that the features contain a direct label encoding. Under correction it falls 0.3005 macro-F1, and its interval widens by an order of magnitude relative to the other models (± 0.0089 against ± 0.0009 for the trees)—the behaviour of a model that is genuinely struggling, as it should be.

Honest ceiling. Under correction the best model reaches 0.9503 ± 0.0011 macro-F1—respectable, and roughly four points below the above-99% range the literature reports.

5.2.1. Deep models are equally deceived

A reviewer may reasonably object that the literature’s headline numbers come from deep architectures, and that a deep model might have been robust to the artifact. It is not. Table 5 trains a deep MLP (61,378 parameters under the distributed recipe, 59,586 under the corrected one, the difference being input width) and a 1D-CNN (6,658 parameters)—the latter matching the construction used in several published Edge-IIoTset papers—under both protocols on a single identical stratified split. This is a separate experiment from the fifteen-fold comparison in Table 4, whose MLP is the smaller (64, 32) estimator from the classical suite.

Table 5: Deep baselines under both protocols (ML subset, 15 epochs).

Architecture	Accuracy (A)	Accuracy (B)	Macro-F1 (A)	Macro-F1 (B)	$\Delta F1$
1D-CNN	1.0000	0.9003	1.0000	0.7598	0.2402
MLP	1.0000	0.9093	0.9999	0.7848	0.2151

Both reach 1.0000 accuracy under the distributed recipe, and both lose more than 0.21 macro-F1 under correction—a larger drop than any tree-based model. Depth confers no protection whatsoever. A network consuming a leaked one-hot feature reads it exactly as a decision stump does, and its additional capacity is spent fitting a signal that will not exist at deployment.

5.2.2. The effect holds at scale on the deep-learning subset

Table 6 repeats the protocol comparison on the DNN subset. After cleaning, 1,909,671 rows remain; the corrected protocol rewrote 15,218,483 placeholder cells and removed 309,530 duplicates.

Table 6: Protocol comparison on the DNN subset (1,909,671 rows).

Model	Accuracy (A)	Accuracy (B)	Macro-F1 (A)	Macro-F1 (B)	$\Delta F1$
Hist. gradient boosting	1.0000	0.9694	1.0000	0.9612	0.0388
Random forest	1.0000	0.9596	1.0000	0.9499	0.0501
Decision tree	1.0000	0.9502	1.0000	0.9390	0.0610
MLP	1.0000	0.9141	1.0000	0.8832	0.1168
Logistic regression	1.0000	0.8948	1.0000	0.8562	0.1438
Gaussian naive Bayes	1.0000	0.5203	1.0000	0.5190	0.4810

Every model reaches 1.0000 accuracy on 1.9 million rows under the distributed recipe. The ordering under correction is identical to the ML subset, and naive Bayes again collapses—here to 0.5190 macro-F1, essentially the performance of a coin flip on a task it appeared to have solved perfectly. That the effect reproduces across two subsets differing by an order of magnitude in size, with inverted class balance, rules out sampling accident.

One difference is worth noting: the corrected DNN subset supports slightly *higher* scores than the corrected ML subset (0.9612 against 0.9503 macro-F1 for the best model), the expected consequence of fourteen times more training data. Correcting the artifact does not merely lower scores; it restores the ordinary relationship between dataset size and achievable performance that a saturated benchmark conceals.

5.2.3. The leak is encoding-agnostic

The distributed recipe uses `pd.get_dummies`, and it would be convenient to conclude that only studies adopting that specific call are affected. They are not. The signal is the *distinction between the tokens '0' and '0.0'*; any encoding that preserves that distinction transmits it. Table 7 fits a decision tree to a single column under four encodings in common use.

Across the four leaking columns and all four encodings, the minimum single-column accuracy is 1.0000—the values are not merely similar but identical, because each encoding is a bijection on the token set and a decision tree is invariant to bijective relabelling of a categorical input.

This matters for scoping the problem. The affected population is not “studies that followed `Readme.txt` verbatim” but **studies that treated these columns as categorical at all**—which, given that four of them

Table 7: Single-column held-out accuracy by encoding scheme (ML subset).

Column	One-hot (recipe)	Label / factorize	Ordinal	Frequency
<code>dns.qry.name.len</code>	1.0000	1.0000	1.0000	1.0000
<code>mqtt.conack.flags</code>	1.0000	1.0000	1.0000	1.0000
<code>mqtt.protoname</code>	1.0000	1.0000	1.0000	1.0000
<code>mqtt.topic</code>	1.0000	1.0000	1.0000	1.0000
<code>http.request.version</code>	0.8460	0.8460	0.8460	0.8460
<code>http.request.method</code>	0.8460	0.8460	0.8460	0.8460
<code>http.referer</code>	0.8460	0.8460	0.8460	0.8460

are MQTT and DNS protocol fields that any IIoT feature set would naturally include, is a far larger group. A study reporting label encoding rather than dummy encoding has not thereby avoided the artifact, and cannot be assumed unaffected.

5.3. The curated subsets cannot support agricultural research

Two structural facts emerged during scoping.

Modbus is absent. Across all 157,800 rows of the ML subset, `mbtcp.len`, `mbtcp.trans_id` and `mbtcp.unit_id` are uniformly zero. In the DNN subset, 150 of 2,219,201 rows (0.0068%) carry a non-zero Modbus transaction ID. Any study claiming to model attacks on irrigation actuation using these subsets is not observing Modbus traffic.

Device identity is stripped. The only surviving value of `mqtt.topic` in either subset is `Temperature_and_Humidity` (1,246 rows in ML; 83,016 in DNN). Soil moisture, pH and water level—the sensors that define precision agriculture—carry no topic attribution. Filtering to “agricultural sensors” is therefore impossible on the curated data.

Both are recoverable from the raw captures, which retain 1,192,777 soil-moisture rows, 2,295,288 water-level rows, 1,615,722 temperature-humidity rows, 746,908 pH rows, and 159,502 Modbus rows.

5.4. AgriEdge: construction and validation

Table 8 summarises the rebuilt benchmark: 1,276,122 rows, 49 features, 751,559 normal and 524,563 attack (attack rate 0.4111).

Attack classes range from 60,069 (DDoS-HTTP) down to 1,229 (MITM) and 1,001 (fingerprinting), preserving natural rarity.

Table 9 re-runs the audit on the constructed benchmark. The comparison with Table 1 is the validation of the method.

No column separates the classes. Maximum separation rate falls from 1.0000 to 0.0288; maximum single-column accuracy from 1.0000 to

Table 8: AgriEdge composition.

Layer	Source	Rows
Perception	pH value	150,987
Perception	Soil moisture	150,341
Perception	Water level	150,187
Perception	Temperature & humidity	150,048
Actuation	Modbus	149,996
Adversary	14 attack classes	524,563

Table 9: Leakage audit of AgriEdge, same columns as Table 1.

Column	Separation rate	1-col. accuracy	NMI	Provenance marker
<code>http.request.version</code>	0.0288	0.6179	0.0641	no
<code>http.request.method</code>	0.0288	0.6179	0.0643	no
<code>mqtt.conack.flags</code>	0.0272	0.5889	0.0366	no
<code>mqtt.topic</code>	0.0272	0.5889	0.0349	no
<code>mqtt.protoname</code>	0.0270	0.5889	0.0363	no
<code>dns.qry.name.len</code>	0.0026	0.5891	0.0041	no
<code>http.referer</code>	0.0005	0.5894	0.0012	no

0.6179; maximum NMI from 0.9859 to 0.0643. No provenance markers remain. Uniform parsing eliminates the artifact by construction.

5.5. Detection fails across layers, not across devices

Table 10 evaluates on AgriEdge (400,000-row subsample, 409 encoded features), withholding each of the five devices in turn. For comparison, all six models score between 0.6914 and 0.9988 balanced accuracy on the same data under a conventional random stratified split.

Table 10: Balanced accuracy under leave-one-device-out, every device withheld in turn.

Model	Soil moisture	Water level	pH value	Temp. & humidity	Modbus	Mean	Worst
MLP	0.9997	0.9998	0.9994	0.9121	0.8164	0.9455	0.8164
Hist. gradient boosting	0.9997	0.9999	0.9995	0.9385	0.7464	0.9368	0.7464
Decision tree	0.9997	0.9999	0.9995	0.9161	0.6793	0.9189	0.6793
Random forest	0.9997	0.9998	0.9995	0.9118	0.5083	0.8838	0.5083
Logistic regression	0.9880	0.9885	0.9880	0.9106	0.4877	0.8726	0.4877
Gaussian naive Bayes	0.6153	0.6248	0.6158	0.6022	0.8068	0.6530	0.6022

The full sweep **refines, and partly overturns, the conclusion a single held-out device would have supported.** Had we withheld only Modbus, we would have reported that introducing any unseen sensor collapses detection. That is not what happens.

Perception sensors are mutually substitutable. Withholding soil moisture, water level or pH costs almost nothing: every model except naive Bayes stays in the 0.9880–0.9999 band, and the tree-based models remain at 0.9995 or better. A detector trained on three agricultural sensors transfers to a fourth essentially without loss. Temperature and humidity is slightly harder (0.9106–0.9385), consistent with it being the one device that carries a distinct MQTT topic in the original capture.

The failure is across layers, not across devices. Only Modbus—the sole actuation-layer device, speaking a different protocol—produces collapse: random forest 0.5083, logistic regression 0.4877, straddling the trivial baseline of 0.5. The generalisation boundary in this benchmark is the perception/actuation layer boundary, not device identity.

This is a sharper and more useful claim than the one a single holdout would have licensed. It says where a practitioner may safely extrapolate—across sensors of similar protocol profile—and where they may not—across an architectural layer. It also refines Hakim et al.’s [5] cross-network result by locating the shift that matters: not merely “a different network,” but a different protocol stratum within the same network.

The ranking inversion is real but layer-specific. Gaussian naive Bayes is by a wide margin the worst model on every perception holdout (0.6022–0.6248) and the *second best* on Modbus (0.8068)—the only model that improves under layer shift. Random forest is the mirror image: joint-best on perception holdouts (0.9997) and worst on Modbus (0.5083). Selecting by mean or by random-split performance would pick random forest; selecting for worst-case robustness picks the MLP, which is simultaneously best on mean (0.9455) and best on the hard case (0.8164). A practitioner optimising average performance would not select the model that survives the shift that actually occurs.

Random splitting remains optimistic in the aggregate. Four models reach 0.9988 balanced accuracy under random splitting, above their worst-case LODO performance by 0.18 to 0.49. This is not leakage in the Section 5.1 sense—the audit confirms no column separates the classes—but random splitting places packets from the same capture session and attack burst on both sides of the boundary. Random-split accuracy on AgriEdge should be read as an upper bound.

5.6. Federated learning: heterogeneity is cheap, the uplink is not

Table 11 reports FedAvg over a 28,450-parameter MLP after 20 communication rounds on a 300,000-row sample, under three client constructions.

Table 11: Global model after 20 FedAvg rounds by client construction.

Client construction	Clients	Accuracy	Balanced accuracy	Macro-F1
IID (prior-work baseline)	5	0.9970	0.9968	0.9969
Per-farm	3	0.9971	0.9970	0.9970
Per-device	5	0.9934	0.9939	0.9932

The result is a negative one, and we report it as such. **Heterogeneity costs remarkably little.** Partitioning one client per sensor type—the maximally heterogeneous construction, in which each client observes a single protocol profile—costs only 0.0037 macro-F1 relative to IID. Per-farm partitioning is indistinguishable from IID.

Heterogeneity does slow convergence: the per-device split reaches 0.9932 macro-F1 only at round 19, while the per-farm split passes 0.9950 by round 8 and plateaus at round 12. Under a fixed round budget the gap would be larger. But the common expectation that non-IID partitioning substantially degrades final federated accuracy is not supported here. We caution that this conclusion inherits the limitation of Section 5.5: it is measured under random splitting of the test set.

The binding constraint is the uplink, not the statistics. Table 12 reports the communication cost of the same model against rural backhaul profiles, assuming five clients uploading dense parameter vectors.

Table 12: Uplink cost of federated training (28,450 parameters, 5 clients, 20 rounds).

Backhaul	Uplink (kbit/s)	Seconds per round	Minutes for full training
LoRaWAN SF7	5.47	832.2	277.4
NB-IoT	250	18.2	6.1
LTE Cat-M1	1,000	4.6	1.5
Rural 4G	5,000	0.9	0.3

On LoRaWAN—a common choice precisely because agricultural sites are remote—a single round costs nearly fourteen minutes of pure uplink, and a 20-round training run costs **4.6 hours** of continuous transmission, before accounting for duty-cycle regulations that cap LoRaWAN transmit time far below 100%. Federated learning in this configuration is not deployable over LoRaWAN at all.

This reframes the design problem. The literature’s emphasis on handling non-IID data is, on this benchmark, addressing the smaller of the two obstacles. A farm gateway can tolerate the 0.0037 macro-F1 that heterogeneity costs; it cannot tolerate 4.6 hours of uplink. Work on gradient compression, sparsification, or reduced round counts would buy far more deployability

here than work on heterogeneity-robust aggregation.

5.7. Edge deployment cost

Table 13 reports deployment footprint for the five models plausible on a gateway.

Table 13: On-gateway deployment cost (single-sample inference, 30 timed repeats).

Model	Size (KiB)	Para- meters	Median latency (μ s)	p95 latency (μ s)	Through- put (/s)
Decision tree	20.9	109	40.1	52.1	24,935
Logistic regression	25.1	410	108.9	123.3	9,183
Gaussian naive Bayes	34.5	1,636	108.9	131.0	9,185
Hist. gradient boosting	388.1	—	4,731.9	6,307.6	211
Random forest	3,563.6	45,002	13,738.8	14,215.1	73

The spread is severe among models whose random-split accuracies are statistically indistinguishable. Random forest and decision tree both score 0.9988 balanced accuracy under random splitting, yet random forest is **342 $\times$ slower** and **171 $\times$ larger**. Its 13.7 ms median inference gives a throughput of 73 samples/second—below the aggregate packet rate of even a modest farm deployment, meaning it cannot run inline.

Read together with Table 10, the practical conclusion is sharp. Random forest is simultaneously the most expensive model to deploy *and* the one that fails hardest across the layer boundary. Its apparent 0.9986 macro-F1 under random splitting is the least informative number in this study. The decision tree, at 1/171 the size and 1/342 the latency, is strictly preferable on both axes.

6. Discussion

6.1. What this means for the Edge-IIoTset literature

We do not claim every published result on Edge-IIoTset is invalid. Studies operating on raw captures, using only numeric protocol features, or excluding the four affected columns are unaffected. But the affected recipe is the one the dataset authors distribute, and `mqtt.topic` and `mqtt.protoname` look like exactly the MQTT-layer features an IIoT detector ought to use. We surveyed public implementations and papers for the recipe’s fingerprints; three observations follow.

First, the recipe is reproduced verbatim in public code. The helper function `encode_text_dummy`—an unusual name, traceable to Jeff Heaton’s teaching material and carried into the dataset’s `Readme.txt`—appears in

public Edge-IIoTset implementations together with the exact fifteen-column drop list and the exact seven dummy-encoded columns, including in a CNN-LSTM implementation. Researchers are copying the recipe, not merely reading it.

Second, and more consequentially, **avoiding `get_dummies` does not avoid the artifact**. Section 5.2.3 shows the leak survives label, ordinal and frequency encoding unchanged. A study that reports “categorical variables were converted using label encoding” has not escaped it.

Third, the diagnostic signature is visible in published results. Hasan et al. [3], who report label rather than dummy encoding, publish a confusion matrix in which their best model produces 4,820 true positives, 25,620 true negatives and zero errors of either kind, alongside 99.94% accuracy. Without their exact column list we cannot attribute this with certainty. We observe only that a flawless confusion matrix on real network traffic is what the artifact produces, that their stated encoding preserves it, and that their correlation-based pruning (dropping features correlated above 0.6) would collapse the four mutually-redundant leaking columns to one—which Table 1 shows is sufficient alone.

We raise this not to single out one paper. The recipe was distributed by the dataset’s authors, the affected columns are the natural ones to use, and the artifact is invisible under default pandas type inference. Following the documentation was reasonable. The point is that the resulting numbers cannot be read as detection performance, and that the affected population is far larger than a citation search for the recipe would suggest.

6.2. *Why the artifact survived*

Three factors plausibly explain four years of silence. **Pandas hides it**: default type inference coerces 0 and 0.0 to an identical float, so a researcher who loads the CSV normally and inspects the columns finds them identical, the difference surfacing only when they are read as strings—which is what `get_dummies` effectively does downstream, after the encoding decision is made. **The result looks like success**: a pipeline returning 99.9% accuracy does not prompt debugging, and the incentive to investigate an anomalously good result is weaker than for a bad one. And **the recipe carries the dataset’s authority**, shipping in its own `Readme.txt`, so following it is the conservative choice and deviation would need justifying.

6.3. *Implications for precision-agriculture deployment*

Select on worst-case, not average, cross-layer performance. Table 10 shows random forest is joint-best on perception holdouts and worst on the actua-

tion holdout. Because the perception cases are nearly free for every model, average LODO performance is dominated by the easy cases and hides exactly the failure that matters. Evaluate against a held-out device from a *different architectural layer* before selecting.

A detector validated on sensors does not transfer to actuators. This is the operationally important consequence. A farm that validates its IDS across its soil, water and pH probes will see excellent numbers and may reasonably conclude the detector generalises. Adding a Modbus irrigation controller then places it in the regime where a third of the model suite performs at chance. Actuation-layer traffic needs its own validation, and arguably its own detector.

Prefer simpler and smaller than the literature implies. The decision tree is $342\times$ faster and $171\times$ smaller than the random forest while degrading far less across the layer boundary (0.6066 against 0.3098 macro-F1). The MLP is best on both mean (0.9455) and worst case (0.8164) at moderate cost, and is our recommended default.

Treat MITM detection as an open problem. The class that matters most for falsified telemetry is represented by 1,229 rows. No result in this paper should be read as evidence that MITM detection in agricultural IIoT is solved.

6.4. Generality of the failure mode

The mechanism—a placeholder serialised differently in two branches of a dataset build, then encoded—is not specific to Edge-IIoTset. It requires only that a dataset be assembled by concatenating separately-parsed sources whose separation correlates with the label, which describes a large fraction of security datasets, where benign and malicious traffic are captured in separate sessions. We would expect the audit to find similar artifacts elsewhere, and we release it as a general tool for that reason.

7. Threats to Validity

Construct validity. Our separation-rate measure treats a column as leaking if its tokens are label-pure, and a legitimately predictive feature could be pure without being an artifact. The provenance probe addresses this by testing the narrower hypothesis that the *spelling of an absence marker* carries the signal—a quantity with no network semantics under any reading. All four flagged columns satisfy that stricter test.

Internal validity. Protocols A and B differ in more than placeholder canonicalisation: B also deduplicates and may encode high-cardinality columns structurally. Our strict-threshold ablation isolates the encoding choice and shows it contributes little, and the 0.52% duplicate rate bounds that contribution, so canonicalisation is the dominant effect. We have not run a full factorial ablation of all four corrections.

External validity. The LODO sweep covers all five devices, which resolves what would otherwise be a serious threat: had we withheld only Modbus, we would have concluded that any unseen device breaks detection, which Table 10 shows is false. The remaining limitation is that AgriEdge holds exactly one actuation-layer device, so the perception/actuation boundary is inferred from a single instance. Confirming it is architectural rather than a peculiarity of this Modbus capture needs a testbed with several distinct fieldbus devices.

Cross-dataset generalisation is out of scope, deliberately. Whether detectors trained on AgriEdge transfer to a different IIoT network is the natural next question, and we do not answer it. Edge-IIoTset’s features are Wireshark protocol-field names (`tcp.flags`, `mqtt.topic`); most comparable datasets—CIC-IoT-2023, TON_IoT, WUSTL-IIoT-2021—expose flow-statistical features from CICFlowMeter or Argus. Name-level alignment yields a near-empty intersection, so any cross-dataset number would measure the alignment more than the detector. A defensible result requires a bridging feature space, which recent work [7] treats as a contribution in its own right. We therefore present the LODO sweep as our generalisation evidence: it shows a genuine distribution shift *within* one testbed, across an architectural layer, large enough to invert model rankings. It does not establish behaviour on a different network, and Table 10 should not be read as a substitute for cross-network validation.

Statistical reporting. The headline protocol comparison (Table 4) is reported over $5\text{-fold} \times 3\text{-repeat}$ stratified cross-validation with 95% intervals. The deep baselines (Table 5), the LODO sweep, the federated runs and the edge-cost measurements are single seeded runs; their effects are large relative to the fold-level variance observed in Table 4, but we do not quantify their variance directly. Cross-validation intervals are also not strictly independent across folds within a repeat, so they should be read as descriptive of spread rather than as frequentist guarantees.

8. Reproducibility

All code is released under the MIT licence as the `agriedge` Python package, at <https://github.com/MostafaGalal1/agriedge>, with the audit, both preprocessing protocols, benchmark construction, the federated simulation and the model suite as separate modules, and nine experiment scripts that each write their own result tables.

All randomness is seeded. Every experiment reported here ran on a 12-core Apple M4 Pro with 25 GB of memory; the deep baselines used the integrated GPU through PyTorch’s Metal backend, training in under a minute each. No experiment required datacentre hardware.

Provenance of the audited copy. Because this paper makes a claim about a distribution rather than a model, the copy matters. Edge-IIoTset is published on IEEE DataPort behind a subscription [16], but the DataPort record itself directs readers to a Kaggle repository under the lead author’s account and gives the retrieval path verbatim. That free channel is therefore author-sanctioned rather than a third-party re-upload, serves the same release, and is what the community in practice trains on; the copy audited here came from it, at exactly that path. Three observations argue the artifact is not an accident of repackaging. The `Readme.txt`, documentation PDF and CSVs all carry the authors’ 18 March 2022 timestamps across ten normal-traffic device directories and fourteen attack captures, and the structure matches the dataset paper. The artifact is present in the authors’ own per-device raw captures—normal-traffic writes 0, attack writes 0.0—not only in the merged subsets, and a mirror that re-archives a download does not regenerate two dozen per-device CSVs. And it is visible in the raw CSV text, independent of any parsing library.

We nonetheless encourage readers to verify against their own copy. The repository includes a dependency-free script that does so in one command, reading every field as text so that no type inference can mask the difference:

```
python scripts/verify_placeholder_split.py \  
    ML-EdgeIIoT-dataset.csv
```

9. Conclusion

Edge-IIoTset carries a serialisation artifact that makes file provenance linearly separable from the label. The distributed recipe converts a placeholder-spelling difference between the normal-traffic and attack build branches into features that recover the label exactly. Four of the seven columns the recipe

names separate every row of both curated subsets—157,800 rows in one, 2,219,201 in the other—and under it five of six standard classifiers attain exactly 1.0000 accuracy in every one of fifteen folds and the sixth attains 0.99998, while a 1D-CNN and a deep MLP reach 1.0000 on a single stratified split. Label, ordinal and frequency encoding transmit the leak identically.

We stop short of claiming that any particular published number is an artifact, since preprocessing is rarely reported in enough detail to tell from outside. What we can say is stronger than a suspicion and weaker than an indictment: the recipe the dataset’s own documentation prescribes yields perfect or all-but-perfect classification for every model we tried, the encodings researchers substitute for it do the same, and the affected columns are the ones an IIoT practitioner would naturally reach for. The mechanism is not specific to this dataset either—it needs only that a corpus concatenate separately-parsed sources whose separation tracks the label, which describes much of the benchmark suite. Any result whose preprocessing touched those columns needs re-checking before it can be read as detection performance.

Correcting the artifact costs the strongest model about five points of macro-F1 and costs the weakest nearly a third of its score. Rebuilding the benchmark from raw captures under uniform parsing eliminates the artifact by construction, restores the Modbus traffic and per-device attribution that agricultural research requires, and yields a benchmark on which no column separates the classes above 0.0288.

On that corrected benchmark, a leave-one-device-out sweep locates where detection stops working. Transfer between perception sensors is nearly free. Transfer across the perception/actuation layer is not: withholding the Modbus gateway drops random forest from 0.9988 to 0.5083 balanced accuracy and logistic regression to 0.4877, straddling the trivial baseline, and inverts the model ranking so that optimising average performance selects the model least able to survive the shift. With a $342\times$ spread in inference latency across models of indistinguishable random-split accuracy, the numbers the field currently optimises say little about deployment.

Our federated results sharpen the same point from a different direction. Non-IID partitioning—the difficulty the federated IIoT literature most often sets out to solve—costs at most 0.0037 macro-F1 on this benchmark, while the uplink required to train even a 28,450-parameter model over LoRaWAN costs 4.6 hours of continuous transmission. The obstacle to federated intrusion detection on a real farm is bandwidth, not statistical heterogeneity.

The corrective is inexpensive: canonicalise placeholders before encoding, deduplicate before splitting, evaluate under held-out-domain protocols,

include a weak model as a leakage canary, and report deployment cost alongside accuracy. We release tooling for all five.

9.1. Future work

Three directions follow. **Cross-network transfer** requires a bridging feature space between packet-field and flow-statistical representations, the principal open problem here. **A second actuation-layer device**—Modbus alongside DNP3, BACnet or OPC-UA—would establish whether the boundary belongs to the layer or to this capture. **Auditing other corpora**: the failure mode needs only that a dataset concatenate separately-parsed sources whose separation tracks the label.

Data availability

Edge-IIoTset is publicly available and free for academic use per its own licence [1]. The `agriedge` package, the AgriEdge benchmark construction pipeline, and all experiment scripts and result tables are released under the MIT licence at <https://github.com/MostafaGalal1/agriedge> and archived under the persistent identifier <https://doi.org/10.5281/zenodo.21941210>, which resolves to the most recent release. The constructed benchmark is not carried in the repository, being 38 MB; it is published as a separate archived dataset at <https://doi.org/10.5281/zenodo.21941319>, so that it can be obtained directly without first acquiring the 10 GB source distribution. It is equally regenerable by running `experiments/03_build_agribench.py` against a local copy of Edge-IIoTset.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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